

Group work set 1: Basics

Q1. Imagine a town in the mountains, where the local economy thrives on the production and sale of “widgets”. The townsfolk rely on these widgets for various tasks, from household chores to powering machinery in the local factories. The townsfolk are reluctant to use banking services, so they hide cash inside their houses. One day, a large fire hits the town, burning 50% of the houses in town. In response, the government asks residents to donate money for repairing the harm caused to the city hall, which also burnt down. Alas, very few people donate. Hence, the government decides instead to impose a per-unit tax on the production of widgets to fund the repairs. Meanwhile, a group of young engineers unveils a groundbreaking invention—a new technology that revolutionizes the way widgets are produced. This innovation slashes production costs dramatically, making widget manufacturing more efficient and affordable than ever before.

Explain what is the anticipated effect of these events on the equilibrium price and quantity of widgets in the market. Provide a brief rationale for your answer.

There are three events here:

- 1) The fire has income effects. Because widgets don't seem to be a luxury produce, this reduces demand. [note: an answer speculating that the rebuilding after the fire would require widgets would be accepted here if argued eloquently, but full points require also mentioning the income effects].
- 2) The per-unit tax increases marginal costs, hence decreases supply.
- 3) The technology innovation decreases marginal costs, hence increases supply.

As these are countervailing effects, it depends which one dominates. If effect (1) dominates, the price and quantity both decrease. If effect (2) dominates, the price will increase and the quantity will decrease. If effect (3) dominates, the price will decrease and quantity will increase.

Q2. "GreenTech" is a firm that produces solar panels. GreenTech has been enjoying high profits for the past year and therefore decided to donate money to the election campaign of the minister of finance. However, two changes have occurred in the market: First, a breakthrough in the production of silicon, a key material used in the production of solar panels, has made it cheaper and more readily available. Second, a new government policy has been implemented, which required the installation of solar panels in every governmental office.

How should these events affect the equilibrium price and equilibrium quantity?

First event: decrease in the price of an input = lower production costs = more supply. This decreases price but increases quantity. Second event: governmental policy causes more demand. This increases price and quantity. Hence, quantity increases but the effect on the price is ambiguous.

Q3. Consider a market for coffee with regular supply and demand curves. The equilibrium price is \$3 and the equilibrium quantity is 1,000 units. Suppose that two events occur simultaneously: First, the price of tea increases. Second, a new roasting technology is developed that lowers the cost of producing coffee. According to the model covered in the course, does the equilibrium price increase, decrease, or stay the same? Does the equilibrium quantity increase, decrease, or stay the same? Explain why.

The numbers don't matter here. Tea is a substitute for coffee, hence an increase in its price results in higher demand for coffee (=higher quantity, higher price). The new tech lowers production costs, implying more supply (=higher quantity, lower price). Hence, the quantity increases but the effect on price is ambiguous.

Q4. Maskovid is a firm that produces facial masks during the Covid-19 pandemic. The market is competitive, i.e. the firm is not a monopoly and there are regular supply and demand curves. Assume that the law says that facial masks must initially be worn indoors by every person. Then, one day, the government announces the following change in policy: From now on, wearing masks indoors is no longer mandatory. Furthermore, the government will subsidize the manufacturing of masks, by giving producers 1\$ per mask produced. Discuss what would be the possible effects of this chain of events on (1) *the equilibrium price and quantity of facial masks*, (2) *the consumer surplus and the supplier surplus*, and (3) *efficiency*?

Effect on the price and quantity:

- Demand: While some people may still wear masks voluntarily, the change is likely to reduce demand for masks on the margins.
- Supply: The subsidy should increase the supply.
- Hence, we get less demand and more supply, both of which suggest that the price will decrease. The quantity may or may not change: less demand means less quantity but more supply means more quantity and it's unknown which effect dominates.

Effect on the surplus:

- The decrease in demand due to the new policy reduces the Consumer Surplus ("CS") and the Supplier Surplus ("SS"). However, the subsidy increases both CS and SS. Hence, the overall effect on the total surplus is ambiguous.

Effect on Efficiency:

- The subsidy creates a Deadweight Loss (DWL), which reduces efficiency. However, the subsidy might resolve the problem of the positive externality (sellers not producing enough masks because they don't care about infections). Hence, it could be that the gain outweighs the DWL, or vice versa.